

Computationally Efficient Data Selection in Gaussian Process Regression-Based Particle Filters for Autonomous Navigation

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ABSTRACT

An unmanned aerial vehicle’s ability to carry out a mission effectively depends greatly on the performance of its navigation algorithms, which are usually based on measurements from an inertial measurement unit. Although inertial navigation on its own is autonomous and reliable, it produces imperfect measurements (bias, scale factors, noise, etc.) which lead to an increasing drift in the navigation solution over time. While this is usually enough for short missions, external recalibration measurements and the use of a fusion filter are necessary for longer missions. GNSS measurements are often used because they complement inertial data, but they are easy to jam and are not always accessible. Terrain-aided navigation, which consists in estimating the position and velocity of a vehicle by comparing the measurement taken by an embedded radar altimeter to a known terrain elevation map, has proven to be an excellent alternative when GNSS data is unavailable.

This method presents two main challenges for its implementation. The first one is that the map is inherently highly non-linear, which renders standard navigation algorithms, such as Kalman filters, unable to converge reliably on the true state of the vehicle. To address this issue, more flexible particle filtering techniques [Arulampalam et al., Feb. 2002] are used thanks to their efficiency in scenarios where the navigation equations are non-linear and non-injective. The well known issue of particle degeneracy is solved by using a Regularised particle filter [Musso et al., 2001], which performs kernel density estimation instead of a regular Monte-Carlo integration.

Moreover, terrain-aided navigation requires the extensive knowledge of the vehicle’s environment, which can take a lot of onboard storage space. In particular, if the area covered by the vehicle is very large, it becomes necessary to reduce the sampling density or the quality of the encoding of the map, leading to an inevitable loss of accuracy in the navigation estimation. Using undersampled and scarce data, it is nonetheless possible to reconstruct a model of the overflowed terrain in the vicinity of the trajectory by interpolation. Kriging [Rasmussen and Williams, 2006] is a powerful statistical interpolation method, based on Gaussian processes, which not only predicts the values of the field at unmeasured points, but also quantifies the uncertainty associated with these predictions. This makes it a key tool for dynamically reconstructing local maps of the magnetic field in real time, improving the accuracy of navigation in poorly known or disturbed environments.

Both particle filtering and kriging are powerful techniques that can easily be combined in a flexible autonomous navigation framework as shown by Figure 1, but they come at a certain computational cost. While methods exist to reduce the processing load of particle filtering, this research focuses on kriging, and particularly on data selection. Since the covariance kernel of a Gaussian process can convey a sense of spatiality, the impact of training data is diminished the further away it is from the interpolation

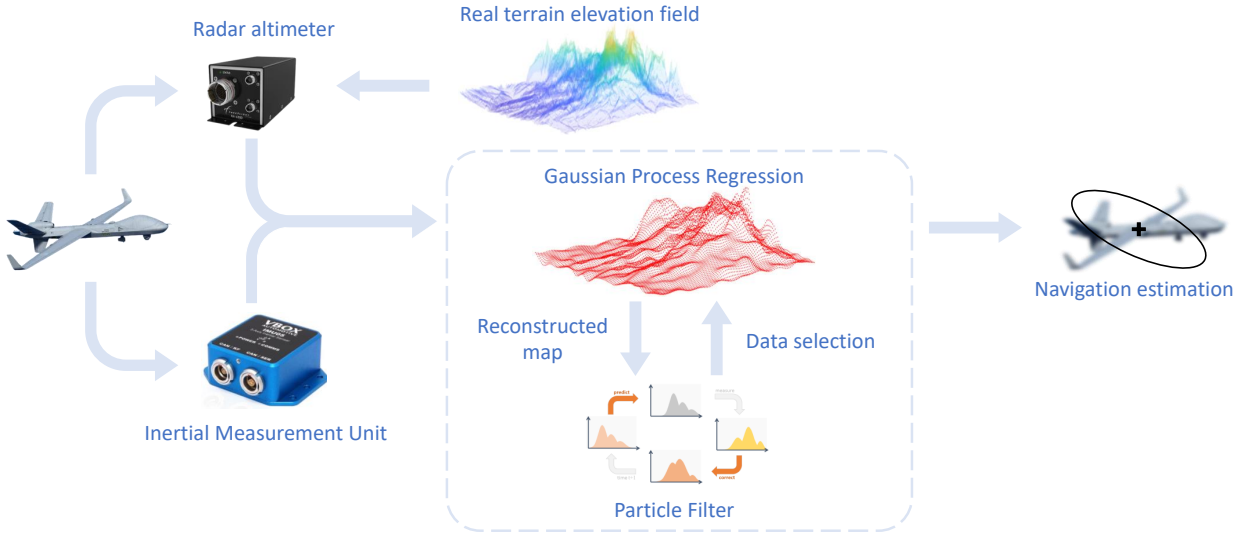


Figure 1: Structure of a kriged terrain-aided particle filter with adaptive data selection

regions, up to a point where they merely slow down the regression without significantly increasing its accuracy. We propose to discuss three dynamical data selection methods, based on the spatial distribution of the filter’s particles, one of which has already been the subject of a conference article [Hubert et al., 2024]. Extensive Monte-Carlo simulations have been performed in a terrain-aided navigation scenario for unmanned aerial vehicle, and show that all three methods improve the real-time performance of the navigation algorithm while maintaining both its accuracy and convergence rate.

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